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Office of Research & Development

VA research on

Traumatic Brain Injury (TBI)

Introduction

The Centers for Disease Control and Prevention (CDC) defines a traumatic brain injury (TBI) as "a disruption in the normal function of the brain that can be caused by a bump, blow, or jolt to the head, or penetrating head injury." Individuals can experience a TBI through everyday activities like playing contact sports, being involved in a car accident, or falling and striking their head. Military service members and Veterans are also at risk of brain injury from explosions experienced during combat or training exercises.

Depending on the severity of the brain injury, a person with TBI may experience a change in consciousness that can range from being dazed and confused to losing consciousness. They may also experience memory loss.

Due to improved diagnostics and increased vigilance, there are now more accurate statistics on military TBI rates. The Defense and Veterans Brain Injury Center (DVBIC) reported nearly 414,000 TBIs among U.S. service members worldwide between 2000 and late 2019. More than 185,000 Veterans who use VA for their health care have been diagnosed with at least one TBI. The majority of those TBIs were classified as mild. TBI and its associated co-morbidities are also a significant cause of disability outside of military settings.

Conditions stemming from TBI can range from headaches, irritability, and sleep disorders to memory problems, slower thinking, and depression. These conditions often lead to long-term mental and physical health problems that can impair Veterans' employment, family relationships, and reintegration into home communities.

The severity of a TBI can usually be assessed through computed tomography (CT) scan (evidence of brain bleeding, bruising, or swelling), the length of loss or alteration of consciousness, the length of memory loss, and how responsive the individual was after the injury. Most TBIs are considered mild, but even mild cases can involve serious long-term effects on areas such as thinking ability, memory, mood, and focus. Other symptoms may include headaches, vision, and hearing problems.

Mild traumatic brain injury (mTBI) is also referred to as a concussion. It can be more difficult to identify than more severe TBI, because there may be no observable head injuries, even on imaging tests, and some of the symptoms may be similar to other problems that stem from combat trauma, such as posttraumatic stress disorder (PTSD).

While most people with mTBI have symptoms that resolve within hours to weeks, a minority may experience persistent symptoms that last for several months or longer. Treatment can include a mix of cognitive, physical, speech, and occupational therapy, along with medication to control specific symptoms such as headaches or anxiety.

A complicating risk factor for mTBI is a person's lifetime accumulation of TBI events. Receiving multiple concussions has been associated with greater risk of developing a neurodegenerative disease like chronic traumatic encephalopathy (CTE). Scientists have found an association between CTE and repetitive mTBI in professional athletes and combat Veterans after autopsy. There is some evidence from epidemiological studies (studies that use clinical diagnostic codes in health records) that shows a link between multiple mTBIs and progressive neurodegenerative conditions, like Parkinson's disease, as well as increased association between the two with increasing severity of TBI.

Brain Games

An arguably unsung hero of his time, VA physician and researcher William Oldendorf developed the prototype for the modern-day CT scan. He could not stand seeing his patients suffer through the top medical recommendations at the time for studying brain injury. Today, VA providers at the Cleveland VA, Drs. Walker and Fu bring their passion and professional expertise

TRAUMATIC BRAIN INJURY FACT SHEET

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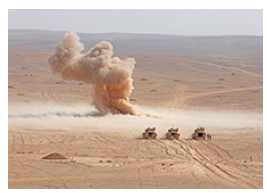

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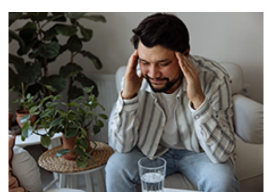

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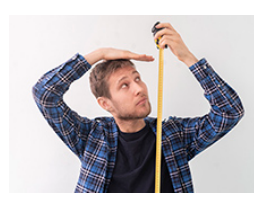

Five ways the Million Veteran Program is transforming our understanding of Veteran health

VA RESEARCH IN THE NEWS

to the realm of virtual reality as they study another unsung struggle: eye focus issues resulting from mild traumatic brain injuries.

From the invention of the CT scan in the 1960s to today's emerging virtual reality, VA has used every technological innovation it can to understand better, study, and treat traumatic brain injuries, or TBIs.

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Selected Major Accomplishments

- **2008:** Established a Brain Bank to collect and study post-mortem human brain and spinal cord tissue to better understand the effects of trauma on the human nervous system
- **2012:**Discovered chronic traumatic encephalopathy, a degenerative disease linked to repeated head traumas, in the brains of four Veterans after their deaths
- **2013:** Along with the Department of Defense, funded two research consortia to improve treatment for PTSD and mTBI: the Consortium to Alleviate PTSD and the Chronic Effects of Neurotrauma Consortium (CENC)
- **2014:** Developed the Boston Assessment of Traumatic Brain Injury—Lifetime (BAT-L), the first post-combat, semi-structured clinical interview to characterize head injuries and diagnose TBIs
- **2015:** Discovered that degeneration of white matter in the brains of Veterans (aging) following exposure to TBI happened independently of concussion symptoms
- **2016:** Identified the cerebellum as particularly vulnerable to repeated blast exposures
- **2017:** Published results from a long-term study (TRACTS) that assessed 850 Iraq/Afghanistan Veterans for the consequences of psychological and brain trauma following TBI
- **2018:**
  - Discovered that functional brain networks that actively interfere with pain perception are disrupted by mTBI in post-9/11 Veterans and service members with and without chronic pain
  - Announced a collaboration with the nonprofit organization PINK Concussions to encourage women to donate their brains upon death for the purpose of research on the effects of TBI and PTSD
- **2019:** Building on work developed by CENC, began the five-year, Long-term Impact of Military Relevant Brain Injury (LIMBIC) study, to better understand the long-term impact of TBI in Veterans

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New, Ongoing, and Published Research

VA research related to TBI is wide-ranging. Researchers are examining various approaches to detect, monitor, and treat Veterans with TBI. Among VA researchers' goals are to shed light on brain changes in TBI, improve screening methods and refine tools for diagnosing the condition, and develop ways to treat brain injury or limit its severity when it first occurs.

In brains of dead athletes, researchers look for clues to head trauma, *News Dio*, Feb. 23, 2020

Still misunderstood: how traumatic brain injuries affect America's Veterans, *Military.com*, Feb. 3, 2020

VA and DoD launch Long-Term Impact of Military-related Brain Injury Consortium (LIMBIC), *Defense News Network*, Oct. 23, 2019

Researchers pinpoint when football players are most likely to develop trauma-induced brain disease, *Fast Company*, Feb. 2, 2020

New research may shed light on combat concussion-dementia link, *Military Times*, Oct. 1, 2019

Could boxing deaths be prevented?, *Medium.com*, Aug. 2, 2019

Brain area tied to emotions is larger in Vets, Service members with mild TBI and PTSD, *Science Daily*, April. 29, 2019

Combat PTSD/TBI increases amygdala size in military patients, *U.S. Medicine*, Sept. 29, 2019

Cyclist Kelly Catlin's family donates her brain for concussion research, *The Washington Post*, March 13, 2019

Combat veterans coming home with CTE, *CBS 60 Minutes*, Sept. 16, 2018

VA AND OTHER U.S. GOVERNMENT ONLINE RESOURCES

VA Translational Research Center for TBI and Stress Disorders

War Related Illness and Injury Study Center, Department of Veterans Affairs

Brain Rehabilitation Resource Center, Department of Veterans Affairs

Polytrauma/TBI System of Care, Department of Veterans Affairs

Understanding Traumatic Brain Injury, Department of Veterans

Researchers are also designing improved methods to assess the effectiveness of treatments and learning the best ways to help family members cope with the effects of TBI and support their loved ones.

VA's TBI Model System (TBIMS), a cross-agency collaboration with the National Institute on Disability, Independent Living, and Rehabilitation Research, is a longitudinal multicenter research program that examines the recovery course and outcomes of Veterans and active duty service members with TBI following comprehensive inpatient rehabilitation. The goal of the system is to conduct research that contributes to evidence-based rehabilitation interventions and practice guidelines that improve the lives of people with TBI.

Improved Understanding of Medical and Psychological Needs in Veterans and Service Members with TBI (IMAP) is an extension of TBIMS. The goals of IMAP are to examine types of long-term physical and psychological health conditions in persons with TBI, the impact of those health conditions on recovery, and chronic rehabilitation needs including accessibility of needed services.

VA's Translational Research Center for TBI and Stress Disorders (TRACTS) aims to better understand the complex problems faced by Iraq and Afghanistan Veterans who have experienced a TBI along with PTSD. TRACTS is co-located at the VA Boston Healthcare System and the Michael E. DeBakey VA Medical Center in Houston. The center is focused on innovations to better diagnose TBI and to develop new treatments that target the combined effects of TBI and stress-related disorders. To that end, VA is conducting a long-term study of 850 Veterans that uses neuroimaging, genetic data, patient histories, and clinical data to develop better treatment options for returning Veterans with TBI and PTSD.

The Brain Rehabilitation Research Center (BRRC) in Gainesville, Florida, is funded by the VA Rehabilitation Research and Development (RR&D) Service. The mission of the BRRC is to develop and test treatments that harness neuroplasticity—the brain's ability to form new neural connections—to substantially improve or restore motor, cognitive, and emotional functions impaired by neurologic disease or injury.

VA's War Related Illness and Injury Study Center, located at the VA medical centers in Palo Alto, California; Washington, D.C.; and East Orange, New Jersey, develops and provides post-deployment health expertise to Veterans and their health care providers through clinical programs, research, education, and risk communication. VA's Office of Public Health directs the center.

VA's Polytrauma/TBI System of Care is an integrated network of specialized rehabilitation programs dedicated to serving Veterans and service members who have experienced TBI along with one or more severe secondary injuries, a condition termed polytrauma. (Click here to see a map of sites within the VHA Polytrauma/TBI system of care.)

The Defense and Veterans Brain Injury Center (DVBIC) is a DOD program serving active duty military, their beneficiaries, and Veterans

Affairs

Health Study for a new generation of U.S. Veterans, Department of Veterans Affairs

Long-term Impact of Military-Relevant Brain Injury Consortium-Chronic Effects of Neurotrauma Consortium (CENC), Department of Defense, Department of Veterans Affairs

Defense and Veterans Brain Injury Center, Department of Defense

NINDS Traumatic Brain Injury Information Page, National Institute of Neurological Disorders and Stroke, National Institutes of Health

Traumatic Brain Injury, Medline Plus, National Institutes of Health

Traumatic Brain Injury & Concussion, Centers for Disease Control and Prevention

Traumatic Brain Injury, Administration for Community Living who have experienced a TBI. The center provides state-of-the-art clinical care, innovative research initiatives, and educational programs. DVBIC supports the five VA Polytrauma Rehabilitation Centers, providing staff and expertise to help individuals with deployment-related TBIs.

For more information on TBI, visit our Depression, Mental Health, Suicide Prevention, Substance Use Disorders, and Posttraumatic Stress Disorder topic pages.

If you are interested in learning about joining a VA-sponsored clinical trial, visit our research study information page.

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➤ The National Research Action Plan

In 2013, VA, DOD, and the Department of Health and Human Services developed a wide-reaching plan to improve access to mental health services for Veterans, service members, and military families.

The National Research Action Plan (NRAP) is improving scientific understanding of TBI, PTSD, various conditions related to both, and suicide. Other goals of the plan include providing effective treatments for these conditions and reducing their occurrence.

Among the highlights of the plan has been the establishment of two joint DOD/VA research consortia. These include the Consortium to Alleviate PTSD, a collaboration led by the University of Texas Health Science Center-San Antonio, and the Long Term Impact of Military-Relevant Brain Injury Consortium-Chronic Effects of Neurotrauma Consortium (LIMBIC-CENC), led by Virginia Commonwealth University.

LIMBIC-CENC serves as the comprehensive research network for DOD and VA that focuses on the long-term effects of mTBI in service members and Veterans. The CENC is designed to conduct basic, clinical, and translational research that seeks to discover interventions and treatments for service members and Veterans who experience chronic effects of TBI.

CENC is working to identify and characterize the biological mechanisms behind chronic injury from TBI and neurodegeneration; investigate the relationship between comorbidities of trauma (psychological, neurological, sensory, motor, pain, cognitive, neuroendocrine) and combat exposure to TBI with neurodegeneration; and assess the effectiveness of treatment and rehabilitation strategies for chronic changes following TBI.

Data from CENC's "Observational Study on Late Neurologic Effects of OEF/OIF/OND Combat" was released to the research community on June 30, 2020. The study enrolled more than 1,700 current and former U.S. service members with varying histories of mTBI, from seven VA medical centers and one military treatment facility. The rich data set for the first 1,550 participants is now available to qualified researchers within the Federal Interagency Traumatic Brain Injury Research Informatics System (FITBIR). The data set contains over 264,000 discrete data points and 5,704 MRI scans.

LIMBIC, which was launched Oct. 1, 2019, is researching mTBIs, and will receive $25 million in funding from DOD and up to $25 million from VA, depending on availability of funds. The program includes both a long-term study of 3,000 to 5,000 Veterans and service members, 80% of whom will have had at least one mTBI. It will also create an epidemiological database of military and VA health records, disability assessments, and all other administrative information on more than 2 million Veterans and service members from U.S. conflicts and wars. Only a small percentage of that cohort has combat-induced brain injuries.

All three federal agencies participating in NRAP are collaborating with academia and not-for-profit foundations. They are standardizing, integrating, and sharing data as appropriate; building new tools and technologies; and maximizing the impact of existing research.

The agencies are working together to explore genetic markers that may demonstrate an association between genes and elevated risk for poorer brain health outcomes. They are also working to identify possible changes in brain circuitry, confirm potential biomarkers for TBI and PTSD, and establish new data-sharing agreements.

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➤ Brain changes in TBI

TBI can result in brain damage that is sometimes subtle. This damage can result in changes in memory, attention, thinking, personality, and behavior that are difficult to diagnose and treat. VA researchers are refining ways to reliably diagnose TBI and to predict Veterans' outcomes and care needs.

Larger amygdalas in Veterans with mTBI and combat-related PTSD—The amygdala is the region of the brain that processes such emotions as fear, anxiety, and aggression. A 2020 study led by researchers at the VA San Diego Health Care System found that Veterans and service members with combat-related PTSD and mild TBI had larger amygdalas than those with only brain injury.

The study included 89 Veterans and active-duty military personnel, about a third of whom had both PTSD and mTBI. The rest had mTBI only and served as a control group. The research team cautioned that their findings don't prove a cause-and-effect relationship, only a correlation. Additional investigation is needed, they concluded, to determine whether amygdala size could be used to screen people at risk for PTSD or whether it could be used to monitor the effectiveness of medical solutions.

Another study published in 2020 by members of the same team found that those with a history of combat-related mTBI have much higher levels of abnormally fast brain waves in two of the four lobes of the cerebral cortex: the pre-frontal and posterior parietal lobes. Those two lobes affect functions including reasoning, organization, planning, execution, attention, and problem-solving. These abnormally fast brain waves could cause poorer cognitive functioning.

Higher psychiatric risk and symptoms—A review of 33 relevant studies, published in 2020 by researchers from the Minneapolis VA Health Care System, found that service members with TBIs have higher rates of PTSD, depressive disorder, substance use disorder, and anxiety disorder than those without TBI. A number of studies from throughout the nation linked TBI to greater severity of PTSD symptoms, and one study showed higher rates of suicide attempts. However, studies on the effect of TBI on the severity of depression and substance use disorder symptoms were mixed.

The researchers believe there needs to be an increased emphasis on the evaluation of psychiatric conditions in service members and Veterans with a history of TBI.

Cerebellum issues—A 2016 study by researchers with the VA Puget Sound Health Care System and the University of Washington identified the cerebellum as particularly vulnerable to repeated blast exposures. The cerebellum, an area of the brain, coordinates and regulates muscle activity.

The investigators looked at brain scans from Veterans who had experienced an average of 21 mTBIs each as a result of explosions. The more blasts they were exposed to, the more likely they were to show lower levels of glucose metabolism in the cerebellum. Glucose metabolism is a marker of brain activity.

The research team also created "shock tubes" to test similar blast effects in mice**—**and found that repeated explosions ruptured part of the blood-brain barrier and led to the loss of neurons in the cerebellum. They also revealed the buildup of proteins associated with dementia and Alzheimer's disease.

Chronic traumatic encephalopathy and TBI—In 2012, an international team including researchers from the Boston VA Healthcare System discovered chronic traumatic encephalopathy (CTE) in the brains of four Veterans after their deaths, including three who had survived explosions from improvised explosive devices. The fourth had suffered multiple concussions in and out of service.

CTE is a degenerative disease thought to be linked to repeated head traumas such as concussions and has been identified in the brains of football players who have committed suicide. It is possible that some of the symptoms of PTSD and other mental health conditions in Veterans are caused by CTE.

Another study by the team, published in 2017, found that a protein found in the brain, CCL11, might be used as a biomarker to potentially allow CTE to be diagnosed in living people. By analyzing recently deceased athletes, the team found that those with CTE**—**but not those with dementia or Alzheimer's disease—had significantly elevated levels of CCL11 in their spinal fluid, which creates the possibility that CC11 can also be measured in living people, and a determination can be made whether they have CTE. Previously, CTE could only be correctly diagnosed and distinguished from Alzheimer's disease after death.

VA, along with Boston University and the Concussion Legacy Foundation, have established a Brain Bank to collect and study post-mortem human brain and spinal cord tissue to better understand the effects of trauma on the human nervous system. The Brain Bank has both Veteran and non-Veteran brain samples and has been at the forefront in research neurodegenerative disease that could be linked to lifetime TBI exposure.

Electroencephalograms and mTBI—mTBI and PTSD often cause similar symptoms such as irritability, restlessness, hypersensitivity to stimulation, memory loss, fatigue, and dizziness. In 2016, a team of researchers from the DVBIC used electroencephalograms (EEGs) to learn there were patterns of brain activities at different locations on the scalp for mTBI and PTSD in Iraq and Afghanistan Veterans.

This finding can reduce the possibility these conditions can be confused with each other, thereby improving diagnosis and treatment. It also shows that electrical activity in the brain appears to be affected long after combat-related mTBI, suggesting long term changes in the signaling between cells in the nervous system.

Another study, published in 2017, used a technique called magnetoencephalography to map activity in the brains of patients with PTSD and mTBI. Results indicated that, in PTSD but not mTBI, alpha brain waves showed reductions in network structure (the interconnected symptoms of neurons in the brain). The technique could be another way for clinicians to differentiate between the two conditions.

Tracking mTBI over decades—A federally funded study has enrolled at least 1,200 service members and Veterans who served in Iraq and Afghanistan to learn more about mTBI and how it can best be evaluated and perhaps prevented and treated. The researchers, under the auspices of CENC, hope to follow the cohort for 20 years or more to better understand the long-term neurological effects of mTBI and other deployment-related conditions. Methods and procedures for the study were described in a 2016 article.

The first findings of the study, published in 2018, found that Veterans with mTBI had greater combat exposure; less social support; and more comorbidities, including asthma, PTSD, and sleeping problems. They also had greater pain symptoms and more difficulties in processing information and executive functioning. These findings will guide further research by the study team.

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➤ Effects of TBI

TBI associated with an increased risk of dementia—A 2018 study led by researchers with CENC has found that mTBI with and without loss of consciousness is associated with a heightened risk of developing dementia. The study compared nearly 179,000 Veterans with a TBI diagnosis to a similar group without TBI over a period from

2001 to 2014. In those groups, 2.6% of Veterans without TBI developed dementia, compared with 6.1% of those with TBI.

According to the study team, moderate and severe TBIs had previously been associated with increased dementia risk, but no clear association had previously been demonstrated between dementia and mTBI, especially without loss of consciousness.

The team concluded that even mTBI without loss of consciousness was associated with more than a twofold increase in the risk of a dementia diagnosis. Accordingly, they believe studies of strategies to determine mechanisms, prevention, and treatment of TBI-related dementia in Veterans are urgently needed.

Increased tinnitus risk among service members with TBI— Tinnitus is a condition that involves hearing sound, such as ringing in the ears, when no external sound is present. In a study published in 2019, researchers from the VA San Diego Healthcare System and elsewhere assessed the hearing of 2,600 Marines before and after combat deployment and found that both PTSD and TBI, particularly TBI resulting from a blast, were linked to worsening tinnitus.

Those who already had tinnitus before being deployed found the progression of the condition also increased with hearing loss. The researchers concluded that screening service members before deployment may lead to increased hearing protection for those at risk.

TBI and suicide—Veterans with a history of TBI are more than twice as likely to die by suicide compared to those without such a diagnosis, according to a 2019 study by a team led by researchers from the VHA Rocky Mountain Mental Illness Research Education and Clinical Center in Colorado.

The researchers reviewed electronic medical records of more than 1.4 million Veterans receiving care from VA between Oct. 1, 2005 and Sept. 30, 2015. After adjusting for psychiatric diagnoses such as depression, the team found that those with a moderate or severe TBI were 2.45 times more likely to die by suicide compared to those without a TBI diagnosis. They also found that among those who died, the odds of using firearms as a means of suicide was significantly increased for those with moderate or severe TBI as compared to those without a history of TBI.

Hearing support—A 2015 study by VA's National Center for Rehabilitative Auditory Research (NCRAR) looked at 99 Veterans who were exposed to blasts in Iraq or Afghanistan. All 99 had clinically normal hearing, but all reported problems hearing in difficult listening situations.

The research team asked the Veterans to participate in 10 performance-based tests that have been shown to uncover problems in processing hearing signals. They found that many of the participants had difficulty in one or more of the tests, compared with non–blast-exposed Veterans, although they may have performed well in other tests.

The team concluded that auditory processing symptoms may vary among Veterans exposed to blasts, but that blast injuries can and do result in damage to the central auditory system.

TBI and epilepsy—In 2015, researchers at the VA South Texas Health Care System and the University of Texas reported that Iraq and Afghanistan Veterans with mild TBIs were about 28% more likely to have developed epilepsy than those without TBIs.

The researchers also showed that Veterans who suffered penetrating or severe TBIs had the highest risk of developing epilepsy. The study looked at 256,284 Iraq and Afghanistan Veterans who received either inpatient or outpatient care at VA in 2009 and 2010.

Previous studies of Veterans from World War II and the Korean War have shown a link between combat-related head injury and epilepsy. The research team concluded that because war-related epilepsy in Vietnam Veterans continued 35 years after the war, a detailed, prospective study is needed to understand the long-term relationship between epilepsy and TBI severity in Iraq and Afghanistan Veterans.

TBI and returning to work—Few service members and Veterans with moderate to severe TBI return to work one year after their injury, according to a 2017 VA Polytrauma Rehabilitation Centers study. Those who were older, minorities, or had a more severe TBI were more likely to be unemployed. Of the 293 Veterans and service members in the study, 85% of subjects with severe TBI were unemployed, while 63% with moderate TBI were unemployed. The results will help VA plan rehabilitation services for these patients.

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➤ Diagnosing TBI

Little research on blast versus nonblast TBI—A 2017 review of existing studies by researchers at the Minneapolis VA Health Care System found that little information is available about outcomes for TBIs caused by blasts versus those caused by other factors. The available research showed that blast and nonblast TBI groups had similar rates of depression, sleep disorders, alcohol use, vision loss, balance problems, and functional status.

Results were inconsistent about PTSD, headache, hearing loss, and neurocognitive functions. More research is needed, according to the researchers, on the differences between blast and nonblast TBI, along with consistent definitions of blast exposure.

Post-concussive symptoms after deployment—In a 2017 study by DVBIC, nearly half of soldiers who had an mTBI while serving in Afghanistan or Iraq had post-concussive symptoms such as sleep problems, forgetfulness, irritability, and headaches three months after their deployment. According to the researchers, this suggests that mTBI is associated with continuing problems for longer than has been generally recognized in the active duty population.

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➤ Investigating treatments for TBI

VA researchers are studying the effectiveness of various existing and potential treatments for TBI and symptoms such as headaches, anxiety, and mood swings.

Rehabilitative services—Many Veterans with moderate to severe TBI require rehabilitative services in the areas of engaging in recreation, solving problems, getting around their communities, improving job skills, and accessing psychological support, even five years or more after their injury. A 2019 study, led by researchers at the James A. Haley Veterans' Hospital in Tampa, Florida, also found that both this group and Veterans with mTBI require help in improving memory and controlling physical symptoms.

According to the researchers, many patients, their caregivers, and their providers believe these rehabilitation services were only relevant in the short term after the injury occurs. The team hopes the study will stimulate discussion on how to best address long term needs.

Lithium—Lithium is a mood stabilizing drug that has been used since the 19^{th} century. In 2017, a team of researchers from the VA Pittsburgh Healthcare System found that lithium modestly improved cognitive performance in rats when they were tested 14 days after receiving a TBI. The research also found that lithium does not reduce motor impairment or brain tissue loss following a TBI. Other VA researchers are studying lithium in a cooperative study (CSP #590) to determine whether the drug can be used to prevent suicide. No results have yet been posted for this study.

Protein treatment may protect against TBI damage—A 2017 study by researchers at the VA Pittsburgh Healthcare System found that treatment with UCH-L1, a protein expressed in high levels in neurons, combined with other proteins, has the potential to improve cognitive function when given weeks or months after a TBI. The protein was injected into mice with TBIs, and those treated with the protein showed better brain function than others who were not. Neuron cell survival was also improved.

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➤ Ongoing large-scale TBI research

VA researchers are continually working to better understand TBI and improve therapies for the condition. Ongoing large-scale initiatives designed to improve the health of Veterans with TBI include the Traumatic Brain Injury Veterans Health Registry and the "New Generation" study.

VA Office of Public Health projects—The TBI Veterans Health Registry, begun in 2009 by VA's Office of Public Health (OPH), is providing military and civilian researchers with data on a large number of well-documented cases of TBI from the wars in Iraq and Afghanistan. The registry helps evaluate and compare different therapeutic options and outcomes; compares war related TBI with TBI in civilian patients; and examines the association of TBI with other medical conditions, including PTSD, depression, memory loss, sensory loss, and seizure.

Genetics and TBI—A team of researchers led by investigators at the VA San Diego Healthcare System is using data from the Million Veteran Program (a national VA research program to learn how genes, lifestyle, and military exposures affect health and illness) to examine the influence of genetic factors and neuroendocrine abnormalities on cognitive and psychiatric outcomes in Veterans with TBI histories. The team hopes to uncover information on how underlying genetic factors contribute to recovery after injury.

Studies lacking on women with TBI—A Washington DC VA Medical Center review, published in 2020, found that female Veterans and service members are not well represented in TBI research. The researchers found 29 studies on Veterans with TBI that included women. They found that few studies focused on gender, and most that did had only a small number of female participants. The researchers concluded that more work is needed on how TBI specifically affects female Veterans and service members.

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More on Our Website

- Lab study suggests way to stem long-term damage after brain injury, *VA Research Currents*, June 11, 2020
- VA's TRACTS program seeking breakthroughs on traumatic brain injury, *VA Research Currents,* March 31, 2020
- Highlights of VA research on brain health, *VA Research Spotlight*, March 18, 2020
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- Comprehending the effects of blasts, *VA Research Currents*, June 27, 2019
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- Can older Vets with TBIs benefit from mobile game apps?, *VA Research Currents*, April 18, 2019
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- Iowa study aims to help Vets with TBI struggling with migraines, light sensitivity, *VA Research Currents*, May 31, 2018
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- Studies using electrical stimulation, neuroimaging aim for new insights on TBI, PTSD, *VA Research Currents*, March 28, 2018
- VA HSR&D research topics: traumatic brain injury
- Evidence brief: traumatic brain injury and dementia
- Video: Traumatic Brain Injury

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Selected Scientific Articles by Our Researchers

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